

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Worked Solutions

Fundamentals of Random Variables

Q001 Written

For the sum of two dice, construct the probability distribution and find $P5 \leq X \leq 8$.

Solution

1. List the 36 equally likely pairs, count each sum, and divide each frequency by 36.

Final answer

Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Solution

1. Check that all table probabilities add to 1 before evaluating an interval event.

Final answer

Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Worked Solutions**Fundamentals of Random Variables****Q003 Written**

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Solution

1. A probability distribution is a table of every possible value of X and its probability.

Final answer

Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Solution

1. Build the table, multiply each value by its probability, and add the products.

Final answer

For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.

Worked Solutions

Fundamentals of Random Variables

Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Solution

1. Keep the random variable definition unchanged while forming the table.

Final answer

Use $EX = \sum xPX = x$ after constructing each distribution.

Q006 Written

Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Solution

1. The expected number of successes in repeated identical trials can be checked with n times p .

Final answer

Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Worked Solutions

Fundamentals of Random Variables

Q007 Written

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Solution

1. Constants shift the mean; a multiplier scales the mean by the same factor.

Final answer

Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Solution

1. Do not replace EX^2 by EX^2 ; they are different quantities.

Final answer

Apply linearity for $aX+b$ and use the weighted square sum for X^2 .

Worked Solutions

Fundamentals of Random Variables

Q009 Written

Solve both given Exercise groups of transformed expectations for coins and for balls.**Solution**

1. Write the transformation before substituting the known mean.

Final answer

For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.

Q010 Written

For black balls drawn from the stated box, find EX and VX .**Solution**

1. The distribution is $PX = 0 = 2/7$, $PX = 1 = 4/7$, $PX = 2 = 1/7$. Hence $EX = 6/7$ and $VX = EX^2 - EX^2 = 20/49$.

Final answer

$EX = 6/7$; $VX = 20/49$.

Worked Solutions

Fundamentals of Random Variables

Q011 Written

Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).

Solution

1. Variance measures spread around the mean, not the mean itself.

Final answer

Construct each probability table, then use $VX = EX^2 - EX^2$.

Q012 Written

Find EX and VX for all five given Exercise random variables.

Solution

1. For sampling without replacement, use combinations to obtain the exact probabilities.

Final answer

Use the distribution table for each experiment and the variance identity.

Worked Solutions

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Solution

1. Find VX first, then take the positive square root.

Final answer

The table gives $VX=24/49$, so $\sigma X=2\sqrt{6}/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

Solution

1. Standard deviation has the same units as X and is always non-negative.

Final answer

Use $\sigma X=\sqrt{V(X)}$ after evaluating the variance from the table.

Worked Solutions

Fundamentals of Random Variables

Q015 Written

Find the standard deviation for both given probability-table exercises.

Solution

1. Keep the positive square root only.

Final answer

Compute EX , EX^2 , then $\sigma_X = \sqrt{E(X^2) - E(X)^2}$.

Q016 Written

For the number of winning tickets in the given draw, find VY and σ_Y for $Y=2X+1$ and $Y=-3X-2$.

Solution

1. The constant b changes neither variance nor standard deviation.

Final answer

Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b)=|a|\sigma_X$.

Worked Solutions

Fundamentals of Random Variables

Q017 Written

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X .

Solution

1. Write both transformation rules before substituting numbers.

Final answer

Apply the squared multiplier to VX and the absolute multiplier to σX .

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Solution

1. A translation changes the mean but leaves spread unchanged.

Final answer

Use the variance and standard-deviation transformation formulas for each Y .

Worked Solutions

Fundamentals of Random Variables

Q019 Written

Solve the two given transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Solution

1. The mean equation determines b after the standard-deviation equation fixes a .

Final answer

Use $E(aX + b) = aEX + b$ and $\sigma(aX+b)=|a|\sigma X$, with $a>0$.

Q020 Written

Solve the two given Try problems involving net winnings and determining a, b for a die transformation.

Solution

1. Entry fees are negative constants; they shift the mean only.

Final answer

Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Worked Solutions

Fundamentals of Random Variables

Q021 **Written**

Solve all four given transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Solution

1. Translate the words into $aX+b$ before calculating.

Final answer

Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 **MCQ**

A probability distribution must have probabilities that sum to:

Solution

1. Check the table total.

Correct option: B

Final answer

1

Worked Solutions

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Solution**

1. Only the pair (1,1) gives sum 2.

Correct option: B

Final answer

1/36

Q024 MCQ

The number of heads in three coins can be:**Solution**

1. Count heads in the sample space.

Correct option: B

Final answer

0, 1, 2, 3

Worked Solutions

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Solution**

1. Multiply each value by its probability.

Correct option: B**Final answer**

$$\sum xPX = x$$

Q026 MCQ

The mean number of tails in four fair coins is:**Solution**

1. Use symmetry or the table.

Correct option: B**Final answer**

2

Worked Solutions

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Solution**

1. The values 1 to 6 average to 3.5.

Correct option: B

Final answer

3.5

Q028 MCQ

 $E(aX+b)$ equals:**Solution**

1. Use linearity of expectation.

Correct option: B

Final answer $aEX + b$

Worked Solutions

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Solution**

1. Square values before weighting.

Correct option: C**Final answer**

weighted square sum

Q030 MCQ

If $EX=2$, then $E(3X+2)$ is:**Solution**

1. Calculate $3 \times 2 + 2$.

Correct option: C**Final answer**

8

Worked Solutions

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Solution**

1. Subtract the squared mean.

Correct option: B

Final answer

$$EX^2 - EX^2$$

Q032 MCQ

Variance can never be:**Solution**

1. It is an expected square.

Correct option: C

Final answer

negative

Worked Solutions

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Solution**

1. The table supplies EX and EX^2 .

Correct option: A**Final answer**

draw the distribution

Q034 MCQ

Standard deviation is:**Solution**

1. Take the positive square root.

Correct option: B**Final answer** $\sqrt{V(X)}$

Worked Solutions

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Solution**

1. The square root restores the units.

Correct option: B

Final answer

same units as X

Q036 MCQ

 $V(aX+b)$ equals:**Solution**

1. Constants do not affect spread.

Correct option: B

Final answer

a squared VX

Worked Solutions

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Solution**

1. Standard deviation is non-negative.

Correct option: B**Final answer** $|a| \sigma X$

Q038 MCQ

For $Y=X+3$, the variance is:**Solution**

1. A translation leaves variance unchanged.

Correct option: B**Final answer** VX

Worked Solutions

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Solution**1. The fee is the constant b .

Correct option: B

Final answer

$$aX + b$$

Q040 MCQ

If $a > 0$ and $\sigma(aX + b) = 3$, then a is:**Solution**1. Solve $a \sigma X = 3$.

Correct option: B

Final answer

$$3/\sigma X$$

Worked Solutions

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Solution**

1. b is a translation.

Correct option: C**Final answer**

the mean only

Q042 MCQ

If $EX=2$ and $EX^2=7$, then VX is:**Solution**

1. Calculate $7 - 2^2$.

Correct option: A**Final answer**

3

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 1 Concept Quiz · Written

Find EX and VX for the number of heads in three fair coins.

Solution

1. Use the binomial distribution.

Final answer

$EX=3/2$ and $VX=3/4$

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Solution

1. Use $V(aX + b) = a^2 VX$.

Final answer

$VY=32$ and $\sigma Y=4 \sqrt{2}$

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Solution

1. $EX=7/2$, then use linearity.

Final answer

12

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

Solution

1. Multiply standard deviation by $|a|$.

Final answer

10

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $VX=5$, $V(3X+1)$ is:

Solution

1. Square the multiplier.

Correct option: D

Final answer

45

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For a fair die EX is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

A probability table total must be:

Solution

1. Add all probabilities.

Correct option: B

Final answer

1

Quiz MCQ 8 [Concept Quiz · MCQ](#)

If $\sigma X=4$, $\sigma(X-9)$ is:

Solution

1. Translations do not change spread.

Correct option: B

Final answer

4

Answer key

Use the question number shown on each page.

- | | | | |
|---|---|---|--|
| Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$. | Q009 For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table. | Q019 Use $E(aX + b) = aEX + b$ and $\sigma(aX + b) = a \sigma X$, with $a > 0$. | Q034 B |
| Q002 Use the sample space or combination to list every possible X and sum the relevant probabilities. | Q010 $EX = 6/7$; $VX = 20/49$. | Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules. | Q035 B |
| Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values. | Q011 Construct each probability table, then use $VX = EX^2 - EX^2$. | Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule. | Q036 B |
| Q004 For four fair coins $EX = 2$; for the hypergeometric draw $EX = 4/3$. | Q012 Use the distribution table for each experiment and the variance identity. | Q022 B | Q037 B |
| Q005 Use $EX = \sum xPX = x$ after constructing each distribution. | Q013 The table gives $VX = 24/49$, so $\sigma X = 2\sqrt{6/7}$. | Q023 B | Q038 B |
| Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice. | Q014 Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table. | Q024 B | Q039 B |
| Q007 Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table. | Q015 Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$. | Q025 B | Q040 B |
| Q008 Apply linearity for $aX + b$ and use the weighted square sum for X^2 . | Q016 Use $V(aX + b) = a^2 VX$ and $\sigma(aX + b) = a \sigma X$. | Q026 B | Q041 C |
| | Q017 Apply the squared multiplier to VX and the absolute multiplier to σX . | Q027 B | Q042 A |
| | Q018 Use the variance and standard-deviation transformation formulas for each Y. | Q028 B | Quiz WRITTEN 1 $EX = 3/2$ and $VX = 3/4$ |
| | | Q029 C | Quiz WRITTEN 2 $VY = 32$ and $\sigma Y = 4\sqrt{2}$ |
| | | Q030 C | Quiz WRITTEN 3 12 |
| | | Q031 B | Quiz WRITTEN 4 10 |
| | | Q032 C | Quiz MCQ 5 D |
| | | Q033 A | Quiz MCQ 6 C |
| | | | Quiz MCQ 7 B |
| | | | Quiz MCQ 8 B |